

Sowmya Rajan

Senior Forward Deployed Engineer · Applied AI and Agentic Automation

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SUMMARY

Forward deployed engineer with 5 years at UiPath and a research background in data science. I design and build agentic automation that combines LLM agents, document AI, deterministic automation and human-in-the-loop review, and I work inside customer teams from architecture to deployment. I have delivered for 15+ enterprise customers in insurance, banking, mining, shipping, media and retail. MSc in Digital Sciences, Université Paris Descartes.

SKILLS

AI and LLMs: LLM agents, RAG, Model Context Protocol (MCP), agent memory, human-in-the-loop design, prompt engineering, Claude, OpenAI, Gemini, LangChain, CrewAI, Hugging Face

Document AI and ML: UiPath IXP and Document Understanding, OCR, classification and extraction, computer vision (YOLO, Detectron2), scikit-learn

UiPath platform: Agent Builder, Coded Agents, Maestro (BPMN, Case Management), Data Fabric, Orchestrator, Action Center, Coded Apps, Communications Mining, AI Center, Studio

Engineering: Python, FastAPI, TypeScript, React, Next.js, Node.js, SQL (SQL Server, Snowflake, dbt), REST APIs, SAP integration, Git

EXPERIENCE

Senior Forward Deployed Engineer · UiPath, Bengaluru

Jan 2025 - Present

Design and build agentic automation for enterprise customers, combining AI agents, LLMs, deterministic automation, enterprise systems and human review.

Global media agency · agentic media planning

- Technical advisor to the agency and its implementation partner on an AI system that drafts channel-level media budget plans from a client brief and about nine data sources, then revises them from the planner's plain-language feedback.
- Wrote the proposed architecture, the technical annex and an 18-page plain-language explainer for business and governance stakeholders, and laid out the architecture options the sponsor chose between.

Global mining company · AI work-order automation

- Designed AI triage for SAP maintenance notifications: LLM scoring of description quality, **semantic duplicate detection** with tiered reject, flag and approve thresholds, and a human-in-the-loop supervisor approval step, over English and Spanish data.
- Took three of five process phases through build into UAT across **7 releases**, and built and deployed a React and TypeScript reviewer dashboard for the customer's team.
- Rebuilt the process and solution design documents from code walkthroughs, and led the architecture deep dive and technical review.

Global shipping company · migration recovery

- Recovered a stalled migration between automation platforms for invoicing across **37 countries**, where only 10–14% of 337 machine-translated components had been verified.
- **Fixed 20 defects in 3 weeks** and grouped them into 9 defect classes, including a **silent data-corruption bug with 231 instances** that would have produced wrong invoice amounts without failing tests.

Internal accelerators

- Shipped **8 accelerators in 4 months**, among them IXP Wrapper, which turns 15+ file types into structured JSON and ships as a FastAPI service and a Studio activity, and VisionAI, which compares extraction results across Claude, OpenAI and Gemini.
- Built internal reference implementations of **agent memory** (an agent escalates to a person, then answers repeat requests from memory) and **Maestro case management**, each with a full implementation guide. Delivered two working Hackweek 2026 demos.

Applied Scientist · UiPath, Bengaluru

Nov 2021 – Dec 2024

- Delivered **Document Understanding, Communications Mining and AI Center** solutions for 15+ enterprise customers in insurance, banking, retail, consumer goods, lighting and aviation.
- Built document-vision models for hard extraction cases: signature, seal and table detection (YOLO, Detectron2), handwriting extraction, layout parsing and redaction.
- Built early LLM automation: GPT-based activities and prompts for UiPath workflows, LangChain integrations, coded agents, and an LLM reporting agent for an insurance customer.
- Spoke on Communications Mining and generative AI at UiPath community events, including GenAI Automation Pathshala and the UiPath Adelaide Chapter.

AI Data Specialist, Bixby · Samsung R&D Institute, Bengaluru

Dec 2017 – Aug 2019

- Annotated and curated conversational training data for the NLP models behind **Bixby**, Samsung's virtual assistant, and wrote specifications for new conversational AI capabilities.
- Prepared release builds of NLP models with Python and Bash scripts on the backend.
- Gathered and preprocessed ground-truth image data to train the visual intelligence model behind **Bixby Vision**.
- Received a Spot Award from Samsung R&D Institute for outstanding contribution to vocabulary review.

PROJECTS · GITHUB.COM/SOWMYAARAJAN

Automation Cortex

Python · FastAPI · Next.js · coded agents

Hackweek 2026 build: an intelligence layer over a UiPath automation estate. A static parser turns workflow files into a dependency “genome” for each automation, which drives fragility scores, the blast radius of a proposed change, and generated SOPs. A coded agent challenges design decisions as an adversarial consultant.

FlowLens

TypeScript · Electron · coded agents

Desktop assistant that reads on-screen context such as copied errors and the active app, routes it through a UiPath coded agent, and offers the right automation on a hotkey. It remembers past fixes and spots repetitive work worth automating.

Clausage Bar

Python · Windows tray app

Windows tray monitor for Claude usage: the current session, the weekly window and monthly credits at a glance, with alerts at 50, 75 and 100%. Unofficial, and says so when its data source changes.

HalwaShare

Node.js · WebRTC

Send files of any size, and text, straight to people nearby. Nothing is uploaded or stored: files stream device to device over WebRTC, with a server relay when a direct connection is blocked.

RSNA Pneumonia Detection

Python · YOLO · computer vision

Deep-learning model that detects and localises lung opacities in chest X-rays with bounding boxes.

RESEARCH AND INTERNSHIPS

Web Developer Intern · The Alan Turing Institute

Apr 2021 – Jul 2021

Built the user interface for AutSPACES, an open-source citizen-science platform for the autism community, with Django and Docker. Presented the project at the CC Global Summit 2021.

Cyber Security Researcher Intern · SAP Labs France, Mougins

Nov 2020 – Apr 2021

Retrained NLP models (fastText, Gensim) for SAP Credential Digger, an open-source scanner that finds leaked credentials in GitHub code, and wrote client-side Git hooks that run the scans through Bash.

Research Assistant Intern · Media Neuroscience Lab, UC Santa Barbara, USA

Aug 2020 – Feb 2021

Built NLP pipelines (spaCy, NLTK, pandas) over 42,306 movie plot summaries to link character morality to story performance, and co-submitted an abstract on the work to the 71st ICA Conference.

Data Analyst · Direction du numérique pour l'éducation and CRI, Paris

Feb 2020 – Jul 2020

Analysed real-time logs from the M@gistère platform with Python and SQL to measure how well teachers' online professional development works.

EDUCATION

AIRE Master in Digital Sciences · Université Paris Descartes (now Université Paris Cité), France

2019 – 2021

AI and machine learning track. Awarded a €20,000 EURIP Graduate School scholarship for academic merit.

B.E., Electrical and Electronics Engineering · Visvesvaraya Technological University, India

2013 – 2017